

GMXTRANSPLANT

MD System Builder

Version 0.1.3

Guide to the desktop application and command-line workflows.

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Overview

GMXtransplant builds membrane-protein systems from prepared structures and GROMACS topologies. Give it two CHARMM-GUI folders to transplant proteins and bound ligands automatically, or choose a workflow to insert a protein complex, replace a ligand, restore experimental cholesterol, or place a binder near a membrane protein.

Use the desktop application to select inputs, run the included examples, and inspect results. The gmxtansplant command-line tool supports the same tasks for scripts and remote computers. Each desktop run saves its configuration as run.yaml in the output folder.

All molecular components need compatible parameters before assembly. Inspect the resulting coordinates and topology before simulation. For the two-folder workflow, see the mode guide (CHARMPROT.md), example (examples/charmprot/README.md), and PyMOL/VMD visualization guide (VISUALIZATION.md).

Choose a task

Task	Command option	Example configuration
Replace proteins and ligands using only two CHARMM-GUI folders	--mode charmprot	charmprot.yaml
Insert a protein or protein-ligand complex into a prepared environment	--mode protein	protein_replace.yaml
Replace a bound ligand while retaining the protein	--mode lig	ligand_replace.yaml
Restore experimental cholesterol and adjust leaflet composition	--mode chl	cholesterol_restore.yaml
Add a ligand or protein binder in the water above/below a membrane protein (guide (ADDBINDER.md))	--mode addbinder	addbinder.yaml

Protein mode supports membrane proteins generally, including receptors, channels, and transporters. Any ligands included in the incoming selection move with the protein during alignment.

Install

Use Python 3.10 or newer. On Linux or WSL 2 with WSLg, open a terminal in the downloaded or cloned source directory and run:

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install .
gmxtansplant-gui
```

On macOS, run the same commands in Terminal from the source directory. The command-line tool is gmxtansplant; run gmxtansplant --help to see its options. You can use Conda instead of venv: conda create -n gmxtansplant python=3.12 && conda activate gmxtansplant.

What is installed, and why

Package	Why GMXtransplant needs it
NumPy, SciPy	Coordinate arithmetic, rigid-body fitting and distance searches
PyYAML	Reading and writing the YAML configurations
MDAnalysis	Reading and writing GRO/PDB/MOL2 structures and periodic-box geometry
PySide6 (Qt)	The desktop application's windows. The command-line tool never loads it, so it also works on servers without a display
openbabel-wheel (Open Babel)	The obabel command that adds hydrogens and CHARMM36 names to experimental cholesterol (cholesterol restoration with convert_to_charmm36: t

Optional, installed separately

Package	Needed for	How to install
OpenMM	Running a prepared minimization folder	python3 -m pip install "[minimize]" or conda install
PyMOL (with pymol2)	The pymol_align / pymol_cealign alignment methods; the default mask_fit does not need it	conda install -c conda-forge pymol-open-source
VMD	The Open in VMD comparison view	From the VMD website

On macOS, PyMOL and VMD installed as applications in /Applications are found automatically. For a copy elsewhere, set GMXTRANSPLANT_PYMOL or GMXTRANSPLANT_VMD to its full path.

If a dependency cannot be installed

- Qt on older Linux systems or some clusters: install just the command-line tool with `GMXTRANSPLANT_CLI_ONLY=1 python3 -m pip install ..`. Every mode, example and option remains available through `gmxtransplant`; the application can be added later with `python3 -m pip install "PySide6>=6.6,<7"`.

- Open Babel on Linux or WSL: if the pip package fails, install `GMXtransplant` with `GMXTRANSPLANT_NO_OPENBABEL=1 python3 -m pip install ..`, then install Open Babel with your package manager, such as `sudo apt-get install openbabel`.

- Open Babel on macOS 12 or 13: no ready-built pip package exists, so it is skipped. Install it with `brew install open-babel` or `conda install -c conda-forge openbabel`.

Only cholesterol restoration needs Open Babel; check with `obabel -V`.

For an editable installation, use `python3 -m pip install -e ..`. To run the command line from source without installing the package, use `python3 run_pipeline.py` in place of `gmxtransplant`.

Desktop application

Launch the application from any directory:

```
gmxtransplant-gui
# Equivalent module entry point:
python -m gmxtransplant.gui
```

The desktop application supports Linux, WSL 2 with WSLg, and macOS. It includes an input editor for all configuration sections, file and folder browsing, one-click examples, validation, readable progress summaries, cancellation, and result browsing. Detailed logs are collapsed until View Command Progress is clicked. YAML is generated automatically in the background; there is no YAML editing tab. Select an output folder first. Runs write directly to `charmprot/`, `protein/`, `ligand/`, or `cholesterol/`; example runs use `examples/`. Repeating a run replaces its generated products, including `toppar/`, without timestamped run directories.

Complete example inputs are included in the package for offline use. Read the GUI guide (`GUI_USAGE.md`) for a first run, input setup, results, and platform help.

The macOS installation has been tested on a MacBook Air.

Use a native Python installation matching your Mac's architecture. See the platform help (`GUI_USAGE.md#platform-help`) if the window does not open or the cholesterol workflow reports that Open Babel is missing.

Quick start

In the desktop application

1. Run `gmxtransplant-gui` and choose an output folder at the top of the window.
2. Open the Examples tab and click Run example on any card to see a complete run, or choose a mode on the Configuration tab and browse for your own inputs.
3. Click Check configuration, then Run pipeline, and follow the run on Run & results.

On the command line

Work in a separate directory and generate the example for your task:

```
mkdir my_gmxtransplant_job
cd my_gmxtransplant_job
```

```
gmxtransplant --show-example protein > protein_replace.yaml
```

Edit the paths, residue selections, topology sources, and scientific settings. Then validate the configuration and run assembly:

```
gmxtransplant --mode protein -i protein_replace.yaml --dry-run
gmxtransplant --mode protein -i protein_replace.yaml --output results/
```

--dry-run checks the configuration without opening molecular input files. Input identity, topology, and geometry checks happen during the normal run.

--output DIR chooses where the generated files go. Without it they are written to the directory you ran the command from. Either way, the run's closing line names the absolute folder it wrote to.

Use --show-example lig or --show-example chl for the other modes. Add --no-comments to print a compact example with the same active settings.

Source-checkout example configurations are available for protein insertion (examples/protein_insertion/README.md), ligand replacement (examples/ligand_replacement/README.md), and cholesterol restoration (examples/cholesterol_restoration/README.md). The ligand and cholesterol examples contain input folders for your own structures. Their READMEs list the exact filenames and settings to supply.

Prepare your inputs

The target environment can come from CHARMM-GUI, another system builder, or an MD trajectory frame. Supply matching coordinate and GROMACS topology files, including all force-field and molecule definitions. Incoming proteins and ligands also need matching parameters from a compatible force field.

GRO is preferred: it preserves the periodic box and four-character residue names such as POPC and TIP3. If using an MD frame, preserve atom order, whole molecules, and the correct box. Validated periodic processing requires an axis-aligned orthorhombic cell.

Example files group paths under paths: and reuse them as "\${name}". Unknown or duplicate YAML keys are rejected.

Every input path must be written out in full. No input is resolved against the working directory, against the configuration file's folder, or against any shared input root, and a relative input path is rejected with a message naming the offending key. The one exception is the examples shipped with the package: their paths resolve against the examples/ folder installed alongside the code, so they run anywhere without referring to your filesystem.

Generated files are the mirror image: an output name such as step5_input.gro is a plain filename that lands in the run's output folder. That folder is --output DIR when you pass it, and otherwise the directory you ran the command from. Every run ends by printing the absolute path it wrote to.

Protein insertion

Set these together in protein_replace.yaml:

- target_box.protein_mask: the complete old protein/ligand block to remove.
- replacement_structure.protein_mask: the complete incoming block to insert, including any bound ligands.
- alignment: corresponding protein atoms used for fitting.
- replacement_ligands: coordinate residue names and ITP files for incoming ligands; charge and molecule type are read automatically from the ITP. Use [] when there are none. Each ligand may also give forcefield_path (see below).
- topology.protein_toppar_dir and topology.environment_toppar_dir: the protein and retained environment parameter sources.

The alignment transform is applied to the whole incoming block. Fit a stable protein region so that a changed

ligand pose does not bias placement. For example, a receptor complex can be aligned using corresponding receptor backbone atoms.

mask_fit requires matching atom-name order and residue grouping, equal atom counts, and at least three non-collinear points. Use PyMOL alignment when correspondence must be established between different protein sequences.

Ligand replacement

In ligand_replace.yaml, select the prepared system and the original ligand residue. Use RESNAME:N to select a specific occurrence when its name appears more than once.

The incoming ligand needs three files: its coordinates (new_ligand.coord_path, e.g. a .mol2), its ITP (new_ligand.itp_path), and the force field holding its bond, angle and dihedral parameters (new_ligand.forcefield_path). A CHARMM-GUI ligand ITP only lists which atoms are bonded; the parameters are in the forcefield.itp CHARMM-GUI generated together with it. When forcefield_path is empty, a forcefield.itp next to the ITP is used, otherwise the environment's. Before writing the topology the run checks that every bonded term of the new ligand has parameters, and stops naming any that are missing.

Placement	Behavior
autofit	Fit matching, uniquely named heavy atoms
pairfit	Fit explicit pairs of atom names
mcsfit	Detect the largest shared substructure automatically (same elements and bonds), fit on its closest symmetric match, and report the pairs
nofit	Keep incoming coordinates that are already in the target frame

The protein is retained and protected from clash-based removal. Protein-ligand contacts are reported for inspection.

Cholesterol restoration

In cholesterol_restore.yaml, the experimental protein is aligned to the target protein and the associated cholesterol coordinates are transferred.

Optional CHARMM36 conversion regenerates hydrogens with Open Babel and validates the result against the included cholesterol reference. The example explains the limits for reconstructing missing heavy atoms.

Leaflet composition settings can remove surplus molecules outside the protected region around the protein. They do not create missing lipids or solvent.

Repacking before MD (repack/)

Issue: inserting the experimental cholesterol removes the lipids that overlap them, and the composition step then trims lipids far from the protein. The restored cholesterol start with few lipid neighbours (0–1 lipid heavy atoms within 5 Å instead of about 45). Released straight into MD they left their experimental poses within 25–50 ps in a D1R test, while the membrane shrank by 20%.

What is written: every run also writes repack/, a system set up for a short GROMACS equilibration (about 1.4 ns) that holds every heavy atom of the protein, the ligands and the restored cholesterol, and leaves all other lipids, water and ions free to fill the gaps. It has its own topol.top, toppar/, step5_input.gro, index.ndx and mdp files. A sample run script (sample_run.sh) is also saved. Afterwards, continue from the last GRO with the main topol.top.

Stage	Length	Protein, ligands, restored cholesterol (kJ/mol/nm ²)	Other lipids
step6.0 minimization	5000 steps	4000	free
step6.1 NVT	125 ps	4000	free
step6.2 NVT	125 ps	2000	free
step6.3 NPT	125 ps	1000	free
step6.4 NPT	1 ns	1000	free

The restored cholesterol are their own moleculetype there (toppar/CHLR.itp: CHL1 with all heavy atoms restrained in x, y, z); protein and ligands use their CHARMM-GUI POSRES blocks, and the run warns if a heavy atom has none. In a D1R test with the protein and restored cholesterol held for 1.4 ns, the lipid heavy

atoms within 5 Å of each restored cholesterol rose from 0–1 to 35–56 (normal: 59–73), the gaps closed within about 1 ns, and the poses held. The membrane area keeps settling for 10–15 ns afterwards, so release the remaining restraints gradually. Change the stages, restraints or temperature (310 K) under cholesterol.repack, or set enabled: false.

This restoration option supports cholesterol only. There is currently no corresponding option to preserve and transfer other experimentally resolved bound lipids. Existing lipids in the target membrane are retained according to the clash and composition settings.

Residue selections

Masks use one-based sequential residue positions in coordinate-file order, not the residue numbers printed in PDB or GRO files.

Mask	Selection
:1-298	All atoms in residues 1 through 298
:1,5,10-20	Selected residues or ranges
:1-298@CA	CA atoms in those residues
:1-298@C,CA,N,O	Backbone atoms in those residues
:1-298&!:297-298	Exclude residues 297 and 298

Optional OpenMM minimization

Prepare one openmm_minimization/ folder for a short clash-relaxation step. Preparation does not run or submit minimization. Execute the generated runner separately when ready.

Harmonic positional restraints keep protein and ligand heavy atoms and lipid head groups (the polar P, O, N and S atoms of lipids and sterols) near their starting coordinates. Lipid tails, hydrogens, water, and ions relax freely. These restraints allow small movements; they do not freeze the protein, ligand or membrane.

Prepare the folder

Use case	Command
Assemble and then prepare inputs from that run	gmxtransplant --mode protein -i protein_replace.yaml --prepare-minimization
Prepare inputs from an existing matching system	gmxtransplant prepare-minimization --coordinates step5_input.gro --topology topol.top

--prepare-minimization is an assembly option and works with all four assembly modes. It requires topology.enabled: true and uses the freshly generated GRO and topology. Setting minimization.enabled: true in the assembly YAML has the same effect.

prepare-minimization is a separate command for existing files and does not repeat assembly. Use --include-dir for an additional topology search directory and --output to choose a different destination.

For adjustable standalone settings:

```
gmxtransplant --show-example minimize > minimization.yaml
# Edit the input paths and settings.
gmxtransplant prepare-minimization -i minimization.yaml
```

The folder contains coordinate and topology copies, all topology dependencies, an expanded topology for execution, a settings YAML, the Python runner and its support files, run.sh, generic multicore CPU and GPU SLURM templates, a README, and an input audit with checksums. Move the complete folder to your execution machine; GMXtransplant itself is not required there.

Settings

Every setting has a default suited to CHARMM36 membrane systems, so an assembly configuration needs only minimization: {enabled: true}. The settings you are most likely to change are:

```

minimization:
  enabled: true
  restraint_force_constant_kj_mol_nm2: 1000.0
  max_iterations: 5000
  platform: auto           # or CPU, CUDA, HIP, OpenCL
  restraint_residue_classes: {} # e.g. {MYLIP: lipid} for a custom lipid

```

For nonbonded, constraint, and SLURM resources settings, edit the YAML file.

Restraints use the input coordinates as references and a default force constant of 1000 kJ/mol/nm². Protein molecules are recognized from their amino-acid residues, including attached caps and modifications.

Unrecognized molecules are treated as ligands. Classify custom lipids and unusual residues explicitly, for example `restraint_residue_classes: {MYLIP: lipid}`. Supported classes are protein, ligand, lipid, water, and ion.

The runner uses OpenMM L-BFGS with a finite iteration limit and an RMS objective-gradient tolerance.

Review the example's nonbonded settings for your force field. OpenMM potential switching differs from LJ force switching; this is a configurable clash-relaxation step, not a universal simulation protocol.

Topology position_restraints must be inactive because the runner adds its own restraints. Preparation rejects active definitions instead of ignoring them.

Run separately

From the generated folder:

```

cd openmm_minimization
python3 -m pip install -r requirements.txt
bash run.sh

```

The default run uses multiple CPU threads. Set `OPENMM_CPU_THREADS` to adjust the local thread count, or select a GPU with `bash run.sh --platform CUDA`. HIP and OpenCL can also be selected when available.

For SLURM, review the resource directives and software setup, then submit one template from this folder:

```

sbatch submit_cpu.slurm
# Alternatively:
sbatch submit_gpu.slurm

```

The CPU template uses one process with multiple threads. The GPU template requests one GPU and selects CUDA. An unavailable requested platform produces an error.

Inspect results

The runner writes:

- `results/minimized.gro`
- `results/minimization_report.txt`
- `results/minimization_report.json`

Reports include convergence, energies, forces, selected restraint counts, inferred ligand names, and heavy-atom displacement. Energies include the positional restraints. Exit code 7 means failure or unconfirmed convergence; inspect the report before using any output coordinates.

Existing preparation and results directories are never overwritten. Choose a new destination for preparation, or preserve an earlier results/ directory by renaming it before rerunning. Minimization does not replace equilibration.

Assembly checks and outputs

Assembly checks the periodic environment, aligns the incoming component, handles clashes, optionally corrects charge, and rebuilds the topology molecule list.

Clash cutoffs act on complete environment residues. Every mode uses the same `clash_detection.threshold` of

0.80 Å by default, and no molecule class carries an implicit cutoff of its own: cholesterol is judged exactly like any other lipid. Override a single class through `clash_detection.thresholds` when a system needs it. Classes in `clash_detection.keep_classes` are retained and their contacts are reported. Neutralization removes eligible existing counterions; it does not add ions or set a new salt concentration.

Cholesterol is treated as a distinct molecule only in cholesterol-restoration mode, which is the mode that reasons about it. In protein, ligand and CHARMM-GUI transplant mode it is classified, coloured and reported as an ordinary membrane lipid.

A successful run produces GRO/PDB coordinates, an inspection PDB, text/JSON reports and `index.ndx`, plus `topol.top` and `toppar/` when topology writing is on (as in the examples). Review the alignment, removed molecules, retained clashes, lipid counts, net charge, and coordinate/topology checks before continuing.

The commented example files describe box validation, name restoration for truncated PDB names, clash thresholds, ion protection, and output options.

Limitations

- All components must already have compatible parameters. GMXtransplant does not parameterize new molecules.
- Validated periodic processing supports orthorhombic boxes. Use GRO as the authoritative simulation coordinate output.
- Experimental bound-lipid restoration supports cholesterol only; other bound lipids have no dedicated preservation/restoration option.
- Composition correction removes surplus lipids but does not build missing lipids or solvent.
- Conditional topology definitions may need preprocessing. Minimization preparation supports literal includes, object macros, and `#ifdef/#ifndef` branches; unsupported expressions and cyclic includes are rejected.
- Preparation checks input consistency without importing OpenMM. OpenMM parsing and force-field compatibility are checked when the runner executes.
- Exact overlaps, poor placement, or incorrect parameters may need correction before minimization can succeed.

Development

```
python3 -m unittest discover -v
python3 -m pip install build
python3 -m build
```

GitHub Actions tests Python 3.10, 3.11, and 3.12, builds the package, and runs OpenMM tests with the optional dependency installed.

Runnable protein insertion example

The repository includes a complete input system in `examples/protein_insertion` (`examples/protein_insertion/README.md`). Its relative paths are resolved against the installed example folder, so it runs from anywhere:

```
python -m pip install .
gmxtransplant --mode protein \
-i "$(python3 -c 'from gmxtransplant.gui.model import example_path; print(example_path("protein"))')" \
--output ~/gmxtransplant-example
```

`environment/` supplies the target box, membrane, solvent and ions. `replacement/` supplies the selected incoming protein and bound ligands only. Protein topology definitions come from the `replacement/`; retained environment molecules use environment definitions. Explicit incoming ligand ITPs supersede old copies in

either directory. Conflicts within a source remain errors.

In the generated examples, incoming ligand names and ITP paths are together near the top. `charge: from_itp` and `moleculetype` need not be repeated. Coordinate `resname` and ITP `[ moleculetype ]` names need not be identical: the unique matching `[ atoms ]` residue name determines the type. Ambiguous ITPs still require an explicit molecule type. Detailed YAMLS can set these fields explicitly.

For protein-mode PDB name restoration, the environment's `topol.top` is found beside `environment_toppar_dir`; no separate reference topology is needed. A nonstandard layout can still set `name_restoration.reference_topol` explicitly. The example enables name restoration so truncated POP/TIP/CHL names are classified correctly before clash removal. Set masks, cutoffs, protection, and composition explicitly in YAML. Standalone minimization settings are in `minimization.yaml`.

Desktop application

GMXtransplant helps you build a new molecular system from prepared structures. You choose the inputs, check the setup, run the calculation, and inspect the results in one desktop window. Your input files stay on your computer.

Open the application

If GMXtransplant is already installed, open a terminal and run:

```
gmxttransplant -gui
```

If you have downloaded the source but have not installed it, open a terminal in the GMXtransplant folder and run:

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install .
gmxttransplant -gui
```

The next time you open a terminal, return to the GMXtransplant folder, run `source .venv/bin/activate`, then run `gmxttransplant-gui`. On Linux, use a graphical desktop session. On Windows, use WSL 2 with WSLg. On macOS, open the application from a graphical desktop session. See platform help ([#platform-help](#)) if the window does not open.

Try an included example

The examples include the input structures and settings they need. They are a good way to learn what a completed run looks like.

1. At the top of the window, click Choose folder... and select a folder for results. The box initially shows the folder from which you launched the application; check it before running.

2. Open the Examples tab. Choose an example card and click Run example.

The add binder cards demonstrate a ligand above a receptor and a G protein below it. Other cards cover protein transplant, protein insertion, ligand replacement, and cholesterol restoration.

3. Follow the run in Run & results. Progress at a glance shows the current step and key findings. Click View Command Progress for the full log if you want more detail.

4. When the run finishes, the result folder appears above the file list.

Double-click a report to open it. If PyMOL or VMD is installed, use its button to open the generated view.

The Examples tab

To use an example as a starting point for your own system, click Open in editor. The form will open with its settings filled in. An example run saves to an `examples/` folder; a run started from the editor saves to the selected mode's folder.

Set up your own system

Open the Configuration tab and choose the task that matches your goal:

Task	What it does	Main input
CHARMM-GUI protein transplant	Keeps the membrane, water, and ions from one prepared system, then replaces its protein and ligands with those from another.	Referenc
Protein insertion	Replaces a protein in a prepared system with a new protein and any bound ligands.	Target a
Ligand replacement	Replaces one bound ligand while keeping the rest of the system.	Prepared
Cholesterol restoration	Places experimentally resolved cholesterol into a prepared system.	Target s
Add binder	Places a ligand or protein above or below a membrane protein, with a separate system for each pose.	Prepared
Minimization preparation	Creates a folder of inputs for a restrained OpenMM minimization.	Assemble

Fill in the visible input fields first. Click Browse... or Folder... to select each file or folder. If you type a path, enter its full location. Open Advanced settings when you need to change selections, geometry, topology, composition, or other scientific settings. Help text appears beside each setting. You can switch between tasks without losing the edits you made to each task during this session.

The Configuration tab

Before starting a full run:

1. Click Check configuration to find missing or invalid settings.
2. Click Check input paths to check that the named input files and folders

can be used.

3. Click Run pipeline. The full run also checks the assembled system and reports problems it finds. The two earlier checks do not build a system.

If a check fails, read the message at the top of the window. The Run & results tab has more detail under View Command Progress. Correct the named field and run the check again.

Find and review your results

GMXtransplant creates a subfolder inside the output folder you selected. Each task has its own subfolder, such as protein/, ligand/, cholesterol/, addbinder/, or charmprot/. Included examples go under examples/. An add binder run also creates one folder per pose.

Look for the report and the final coordinate and topology files in that subfolder. run.log contains the detailed messages. run-status.json records whether the run completed, failed, or was cancelled. run.yaml saves the settings and input locations used for that run, so you can keep a record of how the system was built.

Running the same task again in the same output folder replaces files from the previous GUI run. Choose a different output folder if you want to keep both sets of results. Your source inputs are left in place. If an existing file has a name GMXtransplant needs but was not made by an earlier GUI run, the application stops before replacing it. Move that file or choose another output folder.

Click Cancel run to stop a run. A failed or cancelled run may leave partial results in its folder; check the status and log before using those files.

PyMOL and VMD are optional. Their buttons appear only when the application finds them. See the visualization guide (VISUALIZATION.md) for the scene colors and the structures shown together.

Common questions

Why does an input path fail? Use Browse... or Folder... to select the actual input. A typed path must include its full location. Choosing an output folder does not set a base folder for inputs.

Why is a setting missing from the form? Look in Advanced settings on the relevant page. Some settings become visible after you turn on their section.

Where did the output go? Open Run & results. The full result folder is shown above the file list after a run. It is also named at the end of the log.

Why does the cholesterol example mention Open Babel? That workflow needs Open Babel to prepare cholesterol coordinates. If the application reports it missing, install Open Babel and restart GMXtransplant.

Where is the full manual? Click Documentation in the left panel to open the manual included with the application.

You can change the appearance with Theme at the top right. Your choice is remembered the next time you open the application.

Platform help

On Linux, launch the application from a graphical desktop session. If you are connected to a remote computer without a desktop, use the command-line tool described in the README (README.md).

On Windows, use WSL 2 with WSLg and install GMXtransplant inside the Linux environment. Browse using Linux paths, including /mnt/c/ for the Windows C: drive. If the application window does not appear, check that graphical Linux applications work in your WSL installation.

On macOS, install in a Python environment that matches your Mac. The cholesterol workflow also needs Open Babel. Install it separately if GMXtransplant reports that it is missing.

CHARMM-GUI transplant mode

Run the installed package:

```
gmxtransplant --mode charmprot -i charmprot.yaml --dry-run
gmxtransplant --mode charmprot -i charmprot.yaml
```

Or from the source checkout:

```
python -B run_pipeline.py --mode charmprot -i charmprot.yaml --dry-run
python -B run_pipeline.py --mode charmprot -i charmprot.yaml
```

The entire required YAML is:

```
charmprot:
  reference: /path/to/reference/gromacs
  transplant: /path/to/donor/gromacs
```

Each folder must contain `topol.top`, `toppar/`, and `step5_input.gro` (or exactly one `step5*.gro`). Both folder paths, and any `reference_gro/transplant_gro` override, must be written out in full: nothing is resolved against the YAML's directory or the working directory. Inputs are read-only. `output_dir` defaults to the name `charmprot_output`, placed in the `--output` directory when you pass one and otherwise in the directory you ran the command from. Rewriting an existing `output_dir` is the norm: generated artifacts are replaced transactionally, with rollback if publication fails. Unrelated files and GUI logs/locks are preserved. A failed assembly publishes failure reports and removes stale generated results. `--dry-run` validates the YAML without opening the input folders.

Detection and selection

The program reads the actual topology include graph and expands [molecules] using ITP atom counts. Every coordinate atom must agree with its ITP residue and atom name in file order. GRO serial-number wrapping does not affect selection. Unreferenced ITP files do not affect classification.

PROA, PROB, and other PRO... topology molecule types identify proteins. An additional backbone check recognizes protein types with other names. Every detected protein is selected by default. A covalent cofactor inside the same protein moleculetype travels with that molecule; ligands: ignore only applies to separate ligand molecules of the transplant.

Environment categories are water, ion, lipid, sterol, detergent and solvent. The bundled catalogue is a dated snapshot of 943 distinct identifiers from 19 primary-source pages, supplemented with exported water/ion/lipid aliases. It includes every download identifier on the retrieved Individual Lipid Library page and every RESI identifier in the retrieved lipid, water/ion and noble-gas topology files linked from the CHARMM-GUI topology index. It does not treat patch identifiers (PRES) as complete molecules.

Primary sources:

- CHARMM-GUI Individual Lipid Library (<https://www.charmm-gui.org/?doc=archive&lib=lipid>)
- CHARMM-GUI topology index (<https://www.charmm-gui.org/?doc=toppar>)
- Water and ions topology
(https://www.charmm-gui.org/?doc=open_toppar&filename=toppar_water_ions.str)
- Extended ion topology
(https://www.charmm-gui.org/?doc=open_toppar&filename=stream/misc/toppar_ions_won.str)
- CHARMM-GUI water-model FAQ (<https://www.charmm-gui.org/?doc=faq>)
- OpenMM CHARMM water distribution
(<https://github.com/openmm/openmmforcefields/blob/main/charmm/files/waters.yaml>)

The catalogue records a retrieval date, source hashes and per-name source references in `gmxtransplant/charmm_environment.json`. `charmm_environment.py` contains export aliases. Ordinary runs

work entirely offline.

Names are matched exactly, case-insensitively. Both molecule names and the ITP residue names are examined. Thus all catalogued POP... lipids are covered without an unrestricted POP* rule. Single charged atoms also qualify as ions even when their name is custom. Unknown non-protein molecules become inferred ligands and are named explicitly in the report. Topology position after the protein and before the environment is recorded as supporting evidence, not proof of chemical identity.

No fixed name list can guarantee every future, custom or renamed CHARMM-GUI component. The snapshot's coverage is stated precisely above; unfamiliar names remain visible in the report. Use an explicit ligand list or environment override when a name has a different biological role (e.g. a bound metal or functional cholesterol molecule).

Optional YAML

```
charmprot:
  reference: /path/to/reference/gromacs
  transplant: /path/to/donor/gromacs
  reference_proteins: [PROA, PROB]
  transplant_proteins: [PROB, PROA]
  reference_ligands: [LI1]
  transplant_ligands: [LI2]
  output_dir: transplant_result
```

Protein lists specify which complete molecule types to replace and the chain pairing order. Multiple instances of a selected type are all selected, in file order. Omitted protein types in the reference remain in place. Ligand lists take precedence over the environment catalogue; [] explicitly selects no ligands on that side. Omission means automatic detection on that side.

ligands chooses what is inserted from the transplant:

- auto (default): the reference's proteins and ligands are removed, and the

transplant's proteins and ligands are inserted.

- ignore: the reference's proteins and ligands are removed in the same way,

but only the transplant's proteins are inserted; its ligands are left out. reference_ligands may still name which reference ligands to remove; transplant_ligands cannot be combined with ignore.

If a retained protein or ligand (one not selected for removal) clashes with the incoming block, the run stops and reports the conflict.

Additional options:

Option	Default	Meaning
reference_gro, transplant_gro	automatic	Full path to a coordinate file, when the folder's own step5*.gro is not the one to use
environment_overrides	{}	Explicit molecule-name to category mapping, e.g. {MYLIP: lipid}
clash_distance	0.80	Heavy-atom environment clash distance in Å; removes complete molecules
keep_lipids	false	Keep every lipid and sterol, even when it clashes; only other clashing molecules (water, ions, solvent) are removed. Retain
neutralize	true	Remove eligible counterions to restore target charge
target_net_charge	0.0	Desired charge in elementary-charge units
ion_exclusion_distance	10.0	Minimum ion distance from inserted/retained biological molecules in Å
lipid_ion_exclusion_distance	5.0	Minimum ion distance from retained membrane heavy atoms in Å

Alignment, topology and outputs

Corresponding protein chains are globally aligned by residue sequence, followed by a single C α Kabsch rigid fit over all matched chains. This supports missing residues and substitutions without PyMOL. Chain counts must agree; low-identity pairings stop with a clear error. Symmetric chain permutations are not searched automatically: give corresponding protein lists if file order differs. All selected donor atoms, including ligands, receive exactly the same transform. The donor protein should be whole under periodic boundaries before alignment.

Environment clashes use minimum-image distances with the reference box. Whole topology molecules are

removed, including multi-residue lipids. A 0.02 Å margin allows for GRO rounding. Exact molecule charges come from ITP [atoms] records. Charge balancing removes distant counterions with the needed charge; it stops if the requested charge cannot be reached. It does not add ions.

The parameter sets are combined, identical definitions are deduplicated, and conflicting definitions stop the run. Protein and ligand molecule definitions come from the donor; retained molecules keep reference definitions. Conditional include graphs, parameter macros and nested molecule includes need preprocessing; unsupported ambiguity is rejected. Standard flat CHARMM-GUI toppar inputs are the intended format.

On success the output folder contains:

- step5_input.gro, step5_input.pdb: combined coordinates in the reference box
- topol.top, toppar/: assembled parameters and molecule definitions
- index.ndx: topology-based protein, ligand, environment and component groups
- aligned_inspection.pdb: reference and transformed donor selection overlay
- charmprot_report.txt, charmprot_report.json: both systems' classifications,

selection decisions, unknown names, alignment, molecule removals and charge audit

The final topology is reparsed independently and checked against the written GRO atom order and charge. Final files are published together from temporary storage. On failure only diagnostic reports are published, with status: failed. The reports are the record of exactly what was considered environment and ligand.

--prepare-minimization also exports a portable minimization bundle after a successful transplant; it does not run minimization.

GUI, examples and visualization

In the desktop application, select CHARMM-GUI protein transplant and choose the Reference and Transplant folders. Select the output folder at the top of the window. Other settings are under Advanced settings. The Examples card runs the supplied datasets in examples/charmprot/.

Canonical configurations use a charmprot: section and may use the shared paths: registry. The original flat two-folder YAML is still accepted. A root minimization: section uses the same settings as other modes; --prepare-minimization is also supported.

All successful modes produce view.pml, view.vmd, and a visualization/ asset directory. A view.pse session is also created when PyMOL is installed. See Visualization (VISUALIZATION.md) for colors and input overlays.

Comparison views

Each successful assembly exports view.pml, view.vmd, and visualization/ alongside its coordinates. A real view.pse session is generated automatically when PyMOL is installed; PyMOL is found on the PATH, in the running Python environment, or on macOS in /Applications, and GMXTRANSPLANT_PYMOL can point to any other copy. PyMOL and VMD are optional external viewers; neither becomes a CLI or GUI dependency. Minimization preparation shows the prepared input, not an unperformed minimization result.

Use Open in PyMOL or Open in VMD in GUI results, or run from the output folder:

```
pymol view.pml
vmd -e view.vmd
```

Every mode that produces a visualization writes view.vmd and view.pse into its output folder, alongside view.pml and the visualization/ assets, so the assembled system can be reopened later without rerunning the pipeline. view.pse is written during the run when PyMOL is on PATH; when it is not, the run says so and opening view.pml saves view.pse. The PSE is self-contained. When moving a scripted scene, copy the entire output folder, including visualization/, and launch from that folder. VMD locates its assets relative to the .vmd file. For Python-script loading in PyMOL, use run visualization/load.py, module.

Component	Color	Initially visible
Incoming protein	Blue	Yes
Retained protein	Slate	Yes
Incoming ligand	Magenta	Yes
Retained ligand (topology-classified transplant mode)	Purple	Yes
Incoming cholesterol	Gold	Yes
Retained cholesterol	Teal	Yes
Retained membrane	Pale gray	Yes
Original replaced protein	Orange	No
Original replaced ligand	Orange	Yes
Removed environment	Red	No
Reference input	Amber	No
Aligned transplant input	Cyan	No
Water / ions	Pale blue	No

Toggle named objects in PyMOL or molecules in VMD. The old ligand is a comparison overlay, not part of the final molecular system. Full input overlays and removed environment are initially hidden to avoid obscuring the result. visualization/LEGEND.txt and scene.json list the roles and atom counts actually present in that run.

Source overlays use the same rigid transformation as assembly, in the final protein's coordinate frame. Ligand mode shows the positioned incoming ligand; cholesterol mode shows the converted, aligned experimental structure. Box repair, when enabled, is already reflected in the reference overlay. These views do not apply a second fit, wrap molecules, or change scientific coordinates.

Inserted membership comes from the actual inserted atom block, rather than guessing by residue name: an old and a new ligand with the same residue name remain distinct. Two-folder mode additionally uses exact topology molecule categories. Other modes use their standard residue classification for display; unknown retained components are labeled "Other retained molecules." Viewer files are for comparison, not simulation inputs; use the validated final GRO and topology for simulation.

Viewer export failures are reported without invalidating successfully written scientific outputs. Details remain in View Command Progress. The generated scripts work without network access on supported viewer installations on Linux, WSL, and macOS.

Portable and offline use

Install from the source directory:

```
python3 -m pip install .
```

Create and edit a configuration in your job directory:

```
gmxttransplant --show-example protein > protein_replace.yaml
gmxttransplant --mode protein -i protein_replace.yaml --dry-run
gmxttransplant --mode protein -i protein_replace.yaml --prepare-minimization
```

Use modes `lig` or `chl` for ligand replacement or cholesterol restoration. Omit the preparation flag and leave `minimization.enabled: false` for assembly alone.

To prepare an existing matching system:

```
gmxttransplant prepare-minimization --coordinates step5_input.gro --topology topol.top
```

Both workflows create one `openmm_minimization/` folder. Preparation copies and audits the inputs; it does not minimize or submit a job. The folder includes all topology dependencies, the standalone runner, settings, and generic CPU/GPU SLURM templates.

Move the whole folder to your execution machine, activate your Python environment, and review the settings:

```
cd openmm_minimization
python3 -m pip install -r requirements.txt
bash run.sh
```

The runner relaxes clashes with harmonic restraints on protein and ligand heavy atoms and lipid head groups (polar P, O, N and S atoms). Lipid tails, water, ions, and hydrogens relax freely. Classify custom lipids with `minimization.restraint_residue_classes` so they are not treated as ligands.

For SLURM, edit and submit either `submit_cpu.slurm` or `submit_gpu.slurm` from this folder. Inspect `results/minimized.gro` and the convergence reports before continuing. Existing inputs and results are never overwritten.

See `README.md` (`README.md`) for all modes, settings, and limitations.